



Review

Review of computational studies on boiling and condensation



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ABSTRACT

Developments in many modern applications are encountering rapid escalation in heat dissipation, coupled with a need to decrease the size of thermal management hardware. These developments have spurred unprecedented interest in replacing single-phase hardware with boiling and condensation counterparts. While computational methods have shown tremendous success in modeling single-phase systems, their effectiveness with phase change systems is limited mostly to simple configurations. But, given the complexity of phase change phenomena important to many modern applications, there is an urgent need to greatly enhance the capability of computational tools to tackle such phenomena. This article will review the large pool of published papers on computational simulation of boiling and condensation. In the first part of the article, popular two-phase computational schemes will be discussed and contrasted, which will be followed by discussion of the different methods adopted for implementation of interfacial mass, momentum and energy transfer across the liquid-vapor interface. This article will then review papers addressing computational modeling of bubble nucleation, growth and departure, film boiling, flow boiling, and flow condensation, as well as discuss validation of predictions against experimental data. This review will be concluded with identification of future research needs to improve predictive computational capabilities, as well as crucial phase change phenomena found in modern thermal devices and systems that demand extensive computational modeling.

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